

Lecture 4: From Normal Modes to Normalizing Flows

1. Generative Modeling as a Sampling Problem

Many molecular systems are described by high-dimensional probability distributions,

$$p_{\text{data}}(\mathbf{x}), \quad (1)$$

where $\mathbf{x}$ represent atomic coordinates.

For complex molecules, this distribution is typically highly correlated, multimodal, and difficult to sample.

Traditional molecular dynamics generates samples by evolving a trajectory on an energy landscape,

$$\mathbf{x}(0) \rightarrow \mathbf{x}(t_1) \rightarrow \mathbf{x}(t_2) \rightarrow \dots \quad (2)$$

However, if the energy landscape contains large barriers, the trajectory may remain trapped in one metastable basin for a long time.

Generative modeling asks a different question:

Can we construct a simple distribution that is easy to sample,
and learn a transformation from that distribution to molecular data?

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Let the simple distribution be

$$p_z(\mathbf{z}), \quad (4)$$

where the components of $\mathbf{z}$ are often chosen to be independent Gaussian variables,

$$p_z(\mathbf{z}) = \prod_i \frac{1}{\sqrt{2\pi}} e^{-z_i^2/2}. \quad (5)$$

If we can construct a transformation

$$\mathbf{x} = f(\mathbf{z}), \quad (6)$$

then we can generate molecular configurations by

$\mathbf{z} \sim p_z(\mathbf{z}) \longrightarrow \mathbf{x} = f(\mathbf{z}).$

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The challenge is therefore to learn an appropriate transformation between a simple probability distribution and a complex molecular distribution.

2. Probability Transformations Are Familiar in Chemistry

The idea of transforming a complicated distribution into independent variables is not new in chemistry.

A familiar example is normal-mode analysis.

Consider a system with two coupled coordinates,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad (8)$$

and a harmonic potential,

$$U(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T K \mathbf{x}, \quad (9)$$

where

$$K = \begin{pmatrix} k_{11} & k_{12} \\ k_{12} & k_{22} \end{pmatrix}. \quad (10)$$

If

$$k_{12} \neq 0, \quad (11)$$

then the two coordinates are coupled.

The corresponding Boltzmann distribution is

$$p(\mathbf{x}) = \frac{1}{Z} \exp \left[-\frac{\beta}{2} \mathbf{x}^T K \mathbf{x} \right]. \quad (12)$$

Because of the off-diagonal coupling, this distribution does not factorize as

$$p(x_1, x_2) \neq p(x_1)p(x_2). \quad (13)$$

Thus the Cartesian coordinates are correlated.

3. Diagonalizing the Harmonic Potential

Because K is symmetric, it can be diagonalized,

$$K = Q \Lambda Q^T, \quad (14)$$

where

$$Q^T Q = I \quad (15)$$

and

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}. \quad (16)$$

Define the normal-mode coordinates

$$\mathbf{z} = Q^T \mathbf{x}. \quad (17)$$

Equivalently,

$$\mathbf{x} = Q \mathbf{z}. \quad (18)$$

Substituting into the potential,

$$U(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T K \mathbf{x} \quad (19)$$

$$= \frac{1}{2} \mathbf{z}^T Q^T Q \Lambda Q^T Q \mathbf{z} \quad (20)$$

$$= \frac{1}{2} \mathbf{z}^T \Lambda \mathbf{z}. \quad (21)$$

Therefore,

$$U(\mathbf{z}) = \frac{1}{2} \lambda_1 z_1^2 + \frac{1}{2} \lambda_2 z_2^2. \quad (22)$$

The Boltzmann distribution becomes

$$p(\mathbf{z}) \propto \exp\left(-\frac{\beta\lambda_1 z_1^2}{2}\right) \exp\left(-\frac{\beta\lambda_2 z_2^2}{2}\right). \quad (23)$$

Thus,

$$\boxed{p(z_1, z_2) = p(z_1)p(z_2)}. \quad (24)$$

The correlated Cartesian variables have been transformed into independent Gaussian variables.

4. Normal Modes as a Generative Model

Once the normal coordinates are known, sampling becomes easy.

Each normal mode can be sampled independently,

$$z_i \sim \mathcal{N}\left(0, \frac{k_{\text{B}}T}{\lambda_i}\right). \quad (25)$$

The corresponding Cartesian coordinates are then obtained from

$$\mathbf{x} = Q\mathbf{z}. \quad (26)$$

Thus normal-mode analysis provides both directions:

$$\boxed{\mathbf{x} \longleftrightarrow \mathbf{z}}. \quad (27)$$

The forward transformation

$$\mathbf{x} \rightarrow \mathbf{z} \quad (28)$$

removes correlations.

The reverse transformation

$$\mathbf{z} \rightarrow \mathbf{x} \quad (29)$$

generates correlated molecular configurations from independent random variables.

In this sense,

$$\boxed{\text{normal-mode analysis is a linear generative model for a harmonic system.}} \quad (30)$$

5. Normal Modes in Molecular Systems

The same idea applies to larger molecular systems.

Near a local minimum $\mathbf{x}_0$, the potential energy can be expanded as

$$U(\mathbf{x}) \approx U(\mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T H(\mathbf{x} - \mathbf{x}_0), \quad (31)$$

where

$$H_{ij} = \frac{\partial^2 U}{\partial x_i \partial x_j} \quad (32)$$

is the Hessian matrix.

Diagonalizing the mass-weighted Hessian produces molecular normal modes.

For molecules, these correspond approximately to independent vibrational motions.

In crystalline solids, the analogous collective excitations are phonons.

Thus a broad principle appears:

quadratic energy landscape $\implies$ linear transformation $\implies$ independent Gaussian modes.
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6. The Limitation of Normal-Mode Analysis

Normal-mode analysis is exact only for a quadratic potential.

For a complex molecular system,

$$U(\mathbf{x}) \neq \frac{1}{2} \mathbf{x}^T K \mathbf{x}. \quad (34)$$

Biomolecular energy landscapes contain

- anharmonic fluctuations,
- multiple metastable basins,
- conformational transitions,
- nonlinear correlations,
- steric constraints,
- solvent-mediated interactions.

A single linear coordinate transformation cannot generally convert such a distribution into independent Gaussian variables.

Normal-mode analysis is therefore primarily a local approximation around one minimum.

This motivates a more general question:

Can we learn a nonlinear transformation that maps a complex molecular distribution to a simple Gaussian distribution?
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This is the central idea behind normalizing flows.

7. Nonlinear Probability Transformations

Suppose

$$\mathbf{z} \sim p_z(\mathbf{z}) \quad (36)$$

is a simple latent distribution.

We introduce a nonlinear transformation

$$\mathbf{x} = f_\theta(\mathbf{z}), \quad (37)$$

where θ denotes trainable parameters.

If the transformation is invertible,

$$\mathbf{z} = f_\theta^{-1}(\mathbf{x}). \quad (38)$$

Thus we have

$$\boxed{\mathbf{z} \xrightarrow{f_\theta} \mathbf{x}, \quad \mathbf{x} \xrightarrow{f_\theta^{-1}} \mathbf{z}.} \quad (39)$$

The transformation can be interpreted as a nonlinear generalization of the normal-mode transformation.

8. Change of Variables

For a one-dimensional transformation

$$x = f(z), \quad (40)$$

probability conservation requires

$$p_x(x) dx = p_z(z) dz. \quad (41)$$

Therefore,

$$p_x(x) = p_z(z) \left| \frac{dz}{dx} \right|. \quad (42)$$

In many dimensions,

$$\boxed{p_x(\mathbf{x}) = p_z(\mathbf{z}) \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|.} \quad (43)$$

Equivalently, if

$$\mathbf{x} = f_\theta(\mathbf{z}), \quad (44)$$

then

$$p_x(\mathbf{x}) = p_z(\mathbf{z}) \left| \det \frac{\partial f_\theta}{\partial \mathbf{z}} \right|^{-1}. \quad (45)$$

Taking the logarithm,

$$\boxed{\log p_x(\mathbf{x}) = \log p_z(\mathbf{z}) - \log \left| \det \frac{\partial f_\theta}{\partial \mathbf{z}} \right|.} \quad (46)$$

This equation allows the probability density of generated configurations to be evaluated exactly.

9. Normalizing Flows

A normalizing flow is an invertible nonlinear transformation,

$$\mathbf{x} = f_\theta(\mathbf{z}), \quad (47)$$

designed so that

$$p_\theta(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x}). \quad (48)$$

Sampling proceeds as

$$\boxed{\mathbf{z} \sim \mathcal{N}(0, I) \quad \rightarrow \quad \mathbf{x} = f_\theta(\mathbf{z}).} \quad (49)$$

Density evaluation proceeds in the opposite direction,

$$\mathbf{x} \rightarrow \mathbf{z} = f_\theta^{-1}(\mathbf{x}), \quad (50)$$

followed by the change-of-variables formula.

Normalizing flows therefore provide two important capabilities:

$$\boxed{\text{efficient generation} + \text{explicit probability evaluation.}} \quad (51)$$

12. Learning a Normalizing Flow

Because the probability density can be evaluated, a normalizing flow can be trained using maximum likelihood.

Given data samples

$$\mathbf{x}_1, \dots, \mathbf{x}_N, \quad (52)$$

we maximize

$$\mathcal{L}(\theta) = \sum_{n=1}^N \log p_{\theta}(\mathbf{x}_n). \quad (53)$$

Using the change-of-variables formula,

$$\log p_{\theta}(\mathbf{x}) = \log p_z(f_{\theta}^{-1}(\mathbf{x})) + \log \left| \det \frac{\partial f_{\theta}^{-1}}{\partial \mathbf{x}} \right|. \quad (54)$$

Thus training balances two objectives:

- map molecular configurations toward high-probability regions of the latent distribution;
- account correctly for the local compression or expansion of volume.

13. A Broader View: Transporting Probability

The normal-mode and normalizing-flow viewpoints suggest a more general interpretation.

We begin with a simple distribution,

$$p_z(\mathbf{z}), \quad (55)$$

and transport probability into a complicated distribution,

$$p_x(\mathbf{x}). \quad (56)$$

Symbolically,

$$\boxed{p_z \xrightarrow{f_{\theta}} p_{\text{data}}}. \quad (57)$$

The central question in generative modeling is therefore not necessarily how to simulate the original dynamics.

Instead, it is:

$$\boxed{\text{Can we construct an artificial transport process that converts a simple probability distribution into the molecular distribution?}} \quad (58)$$

Normalizing flows answer this question using a deterministic invertible map.

The next lecture will consider a different strategy: stochastic probability transport.